

Supplementary Material: Portfolio-Conditioned Dynamic Weakness and Reinforcement Ranking in Inverter-Rich Power Networks

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1 Policy holdout

The new holdout HARDENING_H01–H24 (N01–N24) is the maximin Latin hypercube of 2000 candidates (seed 20260929; minimum pairwise distance 0.3233 in the unit cube). The file `results/hardening/prereg_inputs/hardening_policies.json` has sha256 53640af64825587f....

Table S1: New holdout policies ($g = u^2$).

id	g	k	t	h	$\alpha_{\perp}(\emptyset)$	base
N01	0.9921	1.264	2.732	1.130	-0.150	stable
N02	0.9036	2.174	0.650	0.703	0.222	unstable
N03	0.4711	1.376	2.531	1.402	-0.153	stable
N04	0.1473	0.701	1.572	0.896	-0.172	stable
N05	0.6857	0.764	2.359	1.509	-0.170	stable
N06	0.8025	0.851	0.991	0.540	-0.125	stable
N07	0.7573	1.979	1.717	0.082	0.005	unstable
N08	0.0628	2.278	1.496	1.000	0.022	unstable
N09	0.0276	1.222	2.104	0.640	-0.167	stable
N10	0.2312	0.877	1.919	1.861	-0.176	stable
N11	0.0001	1.548	1.384	0.251	-0.042	stable
N12	0.1812	1.023	0.862	1.239	-0.109	stable
N13	0.0338	0.598	2.471	0.408	-0.179	stable
N14	0.0066	1.025	2.914	0.219	-0.174	stable
N15	0.0129	1.827	0.792	1.597	0.115	unstable
N16	0.3887	1.893	1.753	0.488	-0.098	stable
N17	0.2655	1.681	1.301	1.808	-0.129	stable
N18	0.1087	1.583	1.977	1.474	-0.165	stable
N19	0.6048	2.073	2.854	1.709	-0.126	stable
N20	0.4438	1.142	1.121	1.952	-0.149	stable
N21	0.2994	0.560	2.592	0.116	-0.178	stable
N22	0.1228	1.759	2.200	0.818	-0.160	stable
N23	0.5486	2.102	0.599	1.317	0.136	unstable
N24	0.0579	1.456	1.164	0.961	-0.084	stable

2 Per-policy census summary

Columns rev./A/B/C: number of units (of nine) with a stable-context reversal (any pair) or a nested reversal at levels A/B/C. The full marginal table (2304 marginals per policy, 63 policies) is `results/hardening/H03_marginals.parquet`; all nested pairs with their curvature witnesses are in `H03_nested_pairs.parquet`.

Table S2: Per-policy reversal counts and oracle fixed-ranking results (FULL stratum).

policy	set	stable base	rev.	A	B	C	C pairs	C s→d/d→s	C max mag	p^*	regret
D01	discovery	y	9	9	8	8	145	143/2	0.045	0.667	0.483
D02	discovery	y	8	8	4	4	65	0/65	0.017	0.722	0.403
D03	discovery	y	8	8	7	7	383	0/383	0.036	0.811	0.351
D04	discovery	y	1	1	0	0	0	0/0	–	0.768	0.164
D05	discovery	y	8	8	7	7	265	204/61	0.041	0.794	0.384
D06	discovery	y	8	8	7	7	420	0/420	0.031	0.786	0.383
D07	discovery	y	8	8	7	7	309	0/309	0.031	0.766	0.348
D08	discovery	y	8	8	6	6	512	511/1	0.037	0.686	0.444
D09	discovery	y	8	8	7	7	491	0/491	0.031	0.801	0.384
D10	discovery	y	9	9	6	6	161	73/88	0.029	0.795	0.360
D11	discovery	y	8	8	5	5	120	1/119	0.036	0.827	0.345
D12	discovery	y	8	8	6	6	162	12/150	0.031	0.797	0.381
D13	discovery	y	8	8	2	2	66	0/66	0.016	0.759	0.355
D14	discovery	y	7	7	4	4	134	0/134	0.038	0.824	0.336
D15	discovery	y	8	8	6	6	206	0/206	0.031	0.775	0.356
H01	old	n	6	6	1	1	3	0/3	0.029	0.726	0.435
H02	old	y	9	9	6	4	5	5/0	0.022	0.638	0.388
H03	old	n	4	4	1	0	0	0/0	–	0.643	0.458
H04	old	y	9	9	2	2	48	6/42	0.025	0.753	0.380
H05	old	y	1	1	0	0	0	0/0	–	0.841	0.152
H06	old	n	7	7	5	5	56	0/56	0.020	0.760	0.304
H07	old	y	8	8	6	6	193	0/193	0.030	0.783	0.380
H08	old	y	8	8	2	2	60	0/60	0.017	0.757	0.366
H09	old	y	9	9	2	2	39	0/39	0.016	0.722	0.402
H10	old	y	8	8	7	7	254	0/254	0.030	0.750	0.358
H11	old	y	9	9	5	5	61	0/61	0.018	0.734	0.358

policy	set	stable base	rev.	A	B	C	C pairs	C s→d/d→s	C max mag	p^*	regret
H12	old	y	9	9	7	7	129	128/1	0.040	0.666	0.425
H13	old	y	1	1	0	0	0	0/0	—	0.733	0.229
H14	old	y	8	8	5	5	174	0/174	0.019	0.784	0.350
H15	old	y	8	8	7	7	294	250/44	0.040	0.815	0.310
H16	old	n	5	5	0	0	0	0/0	—	0.677	0.524
H17	old	n	7	7	1	1	4	0/4	0.013	0.725	0.375
H18	old	y	8	8	7	7	412	0/412	0.027	0.769	0.372
H19	old	y	8	8	5	5	290	0/290	0.038	0.820	0.336
H20	old	n	7	7	4	4	103	0/103	0.033	0.764	0.320
H21	old	y	8	8	3	3	45	0/45	0.017	0.734	0.405
H22	old	y	8	8	7	7	605	0/605	0.032	0.808	0.352
H23	old	y	8	8	6	6	438	429/9	0.035	0.695	0.442
H24	old	y	8	8	8	8	452	408/44	0.047	0.734	0.397
N01	new	y	9	9	4	4	66	0/66	0.016	0.725	0.412
N02	new	n	6	6	0	0	0	0/0	—	0.726	0.452
N03	new	y	9	9	5	5	98	0/98	0.021	0.757	0.353
N04	new	y	8	8	4	4	245	244/1	0.030	0.740	0.380
N05	new	y	8	8	2	2	39	0/39	0.015	0.725	0.410
N06	new	y	8	8	5	5	162	0/162	0.031	0.771	0.363
N07	new	n	8	8	6	6	136	0/136	0.032	0.756	0.305
N08	new	n	8	8	5	5	59	13/46	0.068	0.824	0.287
N09	new	y	9	9	9	7	50	50/0	0.040	0.622	0.422
N10	new	y	8	8	3	3	14	12/2	0.027	0.747	0.379
N11	new	y	3	3	0	0	0	0/0	—	0.775	0.167
N12	new	y	7	7	4	4	103	1/102	0.032	0.825	0.376
N13	new	y	9	9	8	7	87	87/0	0.037	0.654	0.442
N14	new	y	2	2	1	0	0	0/0	—	0.749	0.268
N15	new	n	6	6	3	1	1	1/0	0.028	0.698	0.518
N16	new	y	8	8	7	7	505	0/505	0.030	0.810	0.340
N17	new	y	7	7	6	6	500	0/500	0.036	0.814	0.387
N18	new	y	8	8	7	7	477	474/3	0.031	0.733	0.417
N19	new	y	9	9	6	6	142	0/142	0.022	0.747	0.360
N20	new	y	8	8	6	6	208	0/208	0.030	0.786	0.368
N21	new	y	8	8	2	2	14	7/7	0.025	0.745	0.375
N22	new	y	8	8	5	5	239	124/115	0.040	0.783	0.380
N23	new	n	6	5	0	0	0	0/0	—	0.723	0.595
N24	new	y	8	8	7	7	118	78/40	0.042	0.832	0.325

3 Minimal curvature witnesses

For each new-holdout policy and direction, the level-C nested reversal with the smallest chain length (or, if none has $m = 1$, the largest magnitude), and the certifying second difference d_{ij} on the canonical chain (Proposition 2 of the paper). s→d needs $d > 0$ (a submodularity violation), d→s needs $d < 0$ (a supermodularity violation).

Table S3: Minimal nested EM same-mode reversals and certifying curvature terms (s^{-1}).

policy	dir	i	S_1	S_2	$\Delta(S_1)$	$\Delta(S_2)$	j	d_{ij}	EM-clean
N01	d→s	33	38	34,38	0.015	-0.012	34	-0.027	y
N03	d→s	33	38	34,38	0.021	-0.011	34	-0.032	y
N04	s→d	33	31,34,38	31,34,37,38	-0.021	0.051	37	0.073	y
N04	d→s	34	35,37,38	30,35,37,38	0.030	-0.036	30	-0.066	y
N05	d→s	33	38	34,38	0.012	-0.012	34	-0.025	y
N06	d→s	33	38	34,38	0.022	-0.012	34	-0.033	y
N07	d→s	30	32,34,36	32,34,36,38	0.017	-0.014	38	-0.031	y
N08	s→d	30	32,34,38	32,34,37,38	-0.013	0.038	37	0.051	y
N08	d→s	30	32,34	32,34,38	0.035	-0.013	38	-0.048	y
N09	s→d	32	30,33,35	30,33,35,37	-0.023	0.073	37	0.096	y
N10	s→d	34	31,35,38	31,35,37,38	-0.024	0.031	37	0.055	y
N10	d→s	33	35,38	34,35,38	0.012	-0.011	34	-0.023	y
N12	s→d	31	34,35,38	34,35,37,38	-0.012	0.012	37	0.025	y
N12	d→s	33	31,38	31,34,38	0.038	-0.011	34	-0.049	y
N13	s→d	34	37,38	33,37,38	-0.012	0.015	33	0.027	y
N15	s→d	33	34,38	30,34,38	-0.028	0.039	30	0.067	y
N16	d→s	30	32,34,36	32,34,36,38	0.020	-0.012	38	-0.032	y
N17	d→s	32	30,37	30,33,37	0.011	-0.012	33	-0.023	y
N18	s→d	30	31,35	31,33,35	-0.011	0.039	33	0.050	y
N18	d→s	34	35,37,38	30,35,37,38	0.030	-0.053	30	-0.083	y
N19	d→s	33	38	34,38	0.027	-0.011	34	-0.038	y
N20	d→s	33	38	34,38	0.029	-0.011	34	-0.040	y
N21	s→d	34	31,35,38	31,35,37,38	-0.023	0.029	37	0.051	y
N21	d→s	33	30,38	30,34,38	0.012	-0.012	34	-0.024	y
N22	s→d	33	31,34,38	31,34,37,38	-0.020	0.056	37	0.076	y
N22	d→s	33	31,38	31,34,38	0.013	-0.020	34	-0.033	y
N24	s→d	30	36,38	33,36,38	-0.011	0.203	33	0.214	n
N24	d→s	30	32,38	32,34,38	0.010	-0.026	34	-0.037	y

4 Statistical methodology

Policies and envelope draws are designed points (maximin Latin hypercube; declared bounds), not samples of a physical population. Every fraction is a coverage over the tested conditions. Intervals are percentile bootstraps (10^4 resamples, seed 20260932) with clusters: one per policy; for the 40 fresh draws, one per envelope. Three secondary inferential summaries form the Holm family F_{conf} : T1, an exact one-sided sign test of $p^* < 0.90$ over new base-stable policies; T2, a one-sided cluster sign-flip test of the median paired Spearman advantage minus 0.20; T3, a two-sided permutation test of Spearman($\mu(\theta, \omega_{\text{ref}})$, reversal count). All gates are effect-size thresholds frozen in `docs/CDW_HARDENING_PREREG_V1.md`. Deviations (execution fixes, the ALT-WECC structural-zero and limiter refinements recorded before the cross-model matrix, and the literal application of the GOLD-B eligibility rule) are in `docs/CDW_HARDENING_DEVIATIONS.md`.

5 All branch-ranking baselines

Table S4: Median ranking metrics over 64 conditions: target V_4 , $\gamma = 1.5$, truth $\alpha_\perp$.

predictor	ρ	τ	concord.	top-3	top-5	NDCG@5	sign acc.
S1_absP	0.497	0.389	0.705	0.000	0.200	0.607	-
S2_absS	0.476	0.367	0.692	0.000	0.200	0.607	-
S3_absz	0.237	0.201	0.607	0.000	0.200	0.267	-
S4_elecdist	0.193	0.154	0.583	0.000	0.400	0.376	-
S5_reff	0.300	0.217	0.614	0.333	0.600	0.579	-
S6_fiedler	-0.048	-0.033	0.482	0.333	0.200	0.362	-
S7_betweenness	-0.274	-0.182	0.399	0.000	0.000	0.168	-
S8_dvdq	-0.165	-0.100	0.449	0.000	0.000	0.162	-
S9_dgscr	0.168	0.118	0.567	0.333	0.200	0.300	-
Dconv	0.034	0.011	0.504	0.667	0.600	0.912	0.634
Dfrozen	0.895	0.765	0.897	0.667	0.600	0.968	1.000
Dtotal	0.987	0.929	0.980	1.000	1.000	0.994	1.000
Dport	0.996	0.962	0.999	1.000	1.000	0.999	1.000

Table S5: Median ranking metrics over 64 conditions: target V_4 , $\gamma = 1.10$.

predictor	ρ	τ	concord.	top-3	top-5	NDCG@5	sign acc.
S1_absP	0.495	0.387	0.724	0.333	0.200	0.642	-
S2_absS	0.475	0.360	0.712	0.333	0.200	0.642	-
S3_absz	0.249	0.213	0.642	0.000	0.200	0.260	-
S4_elecdist	0.215	0.173	0.620	0.000	0.400	0.354	-
S5_reff	0.315	0.228	0.654	0.333	0.600	0.577	-
S6_fiedler	-0.006	-0.010	0.508	0.333	0.200	0.357	-
S7_betweenness	-0.247	-0.167	0.385	0.000	0.000	0.157	-
S8_dvdq	-0.145	-0.087	0.454	0.000	0.000	0.128	-
S9_dgscr	0.190	0.139	0.589	0.333	0.200	0.283	-
Dconv	0.060	0.042	0.528	0.333	0.600	0.884	0.556
Dfrozen	0.891	0.752	0.926	0.667	0.600	0.957	1.000
Dtotal	0.999	0.985	1.000	1.000	1.000	1.000	1.000
Dport	0.999	0.989	1.000	1.000	1.000	1.000	1.000

Table S6: Median ranking metrics over 64 conditions: target V_4 , $\gamma = 1.25$.

predictor	ρ	τ	concord.	top-3	top-5	NDCG@5	sign acc.
S1_absP	0.495	0.390	0.713	0.000	0.200	0.629	-
S2_absS	0.475	0.363	0.699	0.000	0.200	0.629	-
S3_absz	0.248	0.213	0.618	0.000	0.200	0.268	-
S4_elecdist	0.209	0.168	0.594	0.000	0.400	0.363	-
S5_reff	0.314	0.224	0.626	0.333	0.600	0.579	-
S6_fiedler	-0.022	-0.018	0.493	0.333	0.200	0.359	-
S7_betweenness	-0.259	-0.172	0.400	0.000	0.000	0.162	-
S8_dvdq	-0.154	-0.089	0.452	0.000	0.000	0.145	-
S9_dgscr	0.185	0.130	0.578	0.333	0.200	0.290	-
Dconv	0.056	0.037	0.518	0.667	0.600	0.897	0.643
Dfrozen	0.894	0.757	0.905	0.667	0.600	0.962	1.000
Dtotal	0.995	0.961	0.998	1.000	1.000	1.000	1.000
Dport	0.998	0.977	1.000	1.000	1.000	1.000	1.000

Table S7: Median ranking metrics over 64 conditions: target V_4 , $\gamma = 1.5$, tracked-mode truth.

predictor	ρ	τ	concord.	top-3	top-5	NDCG@5	sign acc.
S1_absP	0.495	0.389	0.705	0.000	0.200	0.596	-
S2_absS	0.475	0.367	0.692	0.000	0.200	0.596	-
S3_absz	0.237	0.207	0.607	0.000	0.200	0.266	-
S4_elecdist	0.195	0.155	0.583	0.000	0.400	0.376	-
S5_reff	0.303	0.217	0.614	0.333	0.600	0.578	-
S6_fiedler	-0.049	-0.037	0.481	0.333	0.200	0.362	-
S7_betweenness	-0.275	-0.182	0.399	0.000	0.000	0.163	-
S8_dvdq	-0.165	-0.100	0.449	0.000	0.000	0.140	-
S9_dgscr	0.169	0.123	0.572	0.333	0.200	0.300	-
Dconv	0.004	-0.011	0.493	0.667	0.600	0.892	0.611
Dfrozen	0.897	0.768	0.897	0.667	0.600	0.969	1.000
Dtotal	0.988	0.929	0.981	1.000	1.000	0.994	1.000
Dport	0.996	0.962	0.999	1.000	1.000	0.997	1.000

Table S8: Median ranking metrics over 24 conditions: target V_9 , $\gamma = 1.5$.

predictor	ρ	τ	concord.	top-3	top-5	NDCG@5	sign acc.
S1_absP	-0.183	-0.133	0.433	0.000	0.000	0.809	-
S2_absS	-0.196	-0.145	0.427	0.000	0.000	0.809	-
S3_absz	-0.224	-0.159	0.420	0.667	0.400	0.877	-
S4_elecdist	-0.224	-0.167	0.415	0.333	0.200	0.687	-
S5_reff	-0.292	-0.189	0.404	0.667	0.400	0.667	-

predictor	ρ	τ	concord.	top-3	top-5	NDCG@5	sign acc.
S6_fiedler	0.107	0.063	0.532	0.000	0.200	0.742	-
S7_betweenness	0.326	0.222	0.609	0.000	0.200	0.919	-
S8_dvdq	0.171	0.119	0.559	0.333	0.200	0.923	-
S9_dgscr	0.091	0.052	0.526	0.667	0.600	0.953	-
Dconv	-0.195	-0.104	0.448	0.000	0.000	0.307	0.521
Dfrozen	0.943	0.818	0.911	0.667	0.800	0.996	0.927
Dtotal	0.994	0.953	0.978	1.000	1.000	1.000	0.977

6 Cross-model holdout detail

ALT-WECC: TX3 WECC library GFL chain (PLL2, REGCP1, REECB1, REPCA1, BusFreq; frozen parameter set TX3-GFL-0.1, sha256 f07a6a40...) in ANDES 2.0.0 with the TX4 machine transcription. Qualification: all-SG base $\alpha_{\perp}$ reproduced within $1.6\text{e-}07\text{ s}^{-1}$ at all eight policies; descriptor vs ANDES EIG spectra $3.5\text{e-}14$; 488/488 cases initialized. Structural treatment: four decoupled states per converter (REPCA1 s2 integrator, REECB1 Q-PI and V-PI integrators unused under QFLAG=0) are removed exactly before the structural pair (deviation log).

Table S9: Per-policy cross-model results.

policy	ALT base	ALT $\alpha_{\perp}$ (V_4)	Hz	ALT rev.	GFL rev.	τ lines	ALT ρ_{tot}	ALT $\rho_{ P }$	GFL top	ALT top	topo agree
H02	STABLE	0.257	0.472	False	True	-0.030	0.993	0.783	TXother	TXother	1.000
H04	STABLE	0.567	0.399	False	True	0.121	0.986	0.790	TXother	TXother	1.000
H05	STABLE	0.438	0.308	False	False	0.515	0.993	0.790	TXall	TXother	1.000
H07	STABLE	0.397	0.547	False	False	0.091	0.993	0.755	TXother	TXother	1.000
N01	STABLE	0.565	0.417	False	True	-0.121	0.993	0.797	K2_01	TXother	1.000
N02	UNSTABLE	0.440	0.815	False	False	-0.061	0.937	0.371	TXother	TXother	0.667
N03	STABLE	0.594	0.436	False	True	-0.030	0.993	0.783	TXother	TXother	1.000
N04	STABLE	0.371	0.419	False	True	0.061	0.993	0.797	TXother	TXother	1.000

7 Equal-budget corridors

Pending.

8 Modal-mixing decomposition

Two-factor decomposition over 63 policies: controller term variance share 0.19; median $|C|/(|C| + |F|)$ 0.33; decomposition check 0.0e+00. $\mu_{10} = \mu(\theta, \omega_{\text{ref}})$, $\mu_{11} = \mu(\theta, \omega_c(\theta))$ (the earlier campaign's quantity).

Table S10: Mixing correlation tests (primary: new / mu10 / n_rev).

test	ρ	p	Holm p	n
new / mu10 / n_rev	0.548	0.016	None	19
new / mu10 / n_nested	0.548	0.016	0.473	19
new / mu10 / regret	0.274	0.259	1.000	19
new / mu_060 / n_rev	0.548	0.016	0.473	19
new / mu_060 / n_nested	0.548	0.016	0.473	19
new / mu_060 / regret	0.274	0.259	1.000	19
new / mu_070 / n_rev	0.548	0.016	0.473	19
new / mu_070 / n_nested	0.548	0.016	0.473	19
new / mu_070 / regret	0.274	0.259	1.000	19
new / mu_080 / n_rev	0.548	0.016	0.473	19
new / mu_080 / n_nested	0.548	0.016	0.473	19
new / mu_080 / regret	0.274	0.259	1.000	19
new / C / n_rev	0.485	0.036	0.785	19
new / C / n_nested	0.485	0.036	0.785	19
new / C / regret	0.216	0.377	1.000	19
new / mu11 / n_rev	0.483	0.039	None	19
new / mu11 / n_nested	0.483	0.039	None	19
new / mu11 / regret	0.218	0.366	None	19
old / mu10 / n_rev	0.214	0.396	1.000	18
old / mu10 / n_nested	0.214	0.396	1.000	18
old / mu10 / regret	0.133	0.598	1.000	18
old / mu_060 / n_rev	0.214	0.396	1.000	18
old / mu_060 / n_nested	0.214	0.396	1.000	18
old / mu_060 / regret	0.133	0.598	1.000	18
old / mu_070 / n_rev	0.214	0.396	1.000	18
old / mu_070 / n_nested	0.214	0.396	1.000	18
old / mu_070 / regret	0.133	0.598	1.000	18
old / mu_080 / n_rev	0.214	0.396	1.000	18
old / mu_080 / n_nested	0.214	0.396	1.000	18
old / mu_080 / regret	0.133	0.598	1.000	18
old / C / n_rev	0.011	0.968	1.000	18
old / C / n_nested	0.011	0.968	1.000	18
old / C / regret	-0.042	0.867	1.000	18
old / mu11 / n_rev	0.657	0.004	None	18
old / mu11 / n_nested	0.657	0.004	None	18

test	ρ	p	Holm p	n
old / mu11 / regret	0.520	0.029	None	18

9 Reproducibility

Environment: Python 3.13.14, numpy 2.5.2, scipy 1.18.1, pandas 3.0.5 (.venv/tx3-analysis); ANDES 2.0.0 (.venv/xtool-andes-gfl). Preregistration commit 05b507e3. Determinism reruns: identical on every summary field except wall-clock. Orchestrator: experiments/cdw_hardening/CDWH_MASTER_RUN.py; analyses: H03–H19; figures, tables and numbers: CDWH_FIGURES.py, CDWH_TABLES.py, CDWH_NUMBERS.py.

Table S11: Run manifest.

phase	state	n_tasks	n_errors	raw_files	wall_s
H01	COMPLETE	384	0	384	65
H01e	COMPLETE	40	0	40	23
H06	COMPLETE	440	0	440	1286
H08n	COMPLETE	96	0	96	82
H09	COMPLETE	101	0	101	65
H12	COMPLETE	63	0	63	5
H12b	COMPLETE	1	0	1	1
H14	COMPLETE	1312	0	1312	648
H18c	COMPLETE	56	0	56	5
H10	RUNNING	–	–	1965	–
H17/H18 ALT-WECC	COMPLETE	488	0	488	–

10 Supplementary figures

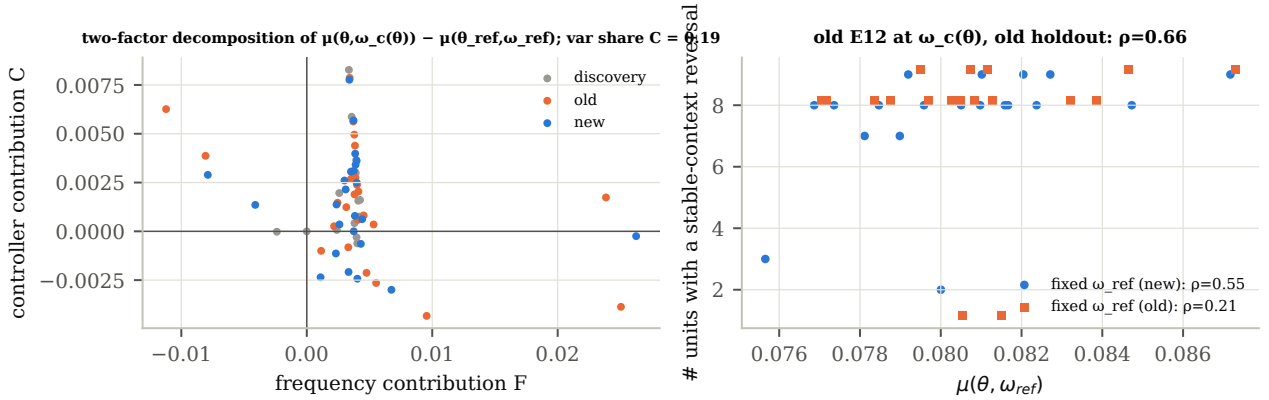


Figure S1: Two-factor modal-mixing decomposition and fixed-frequency correlation.

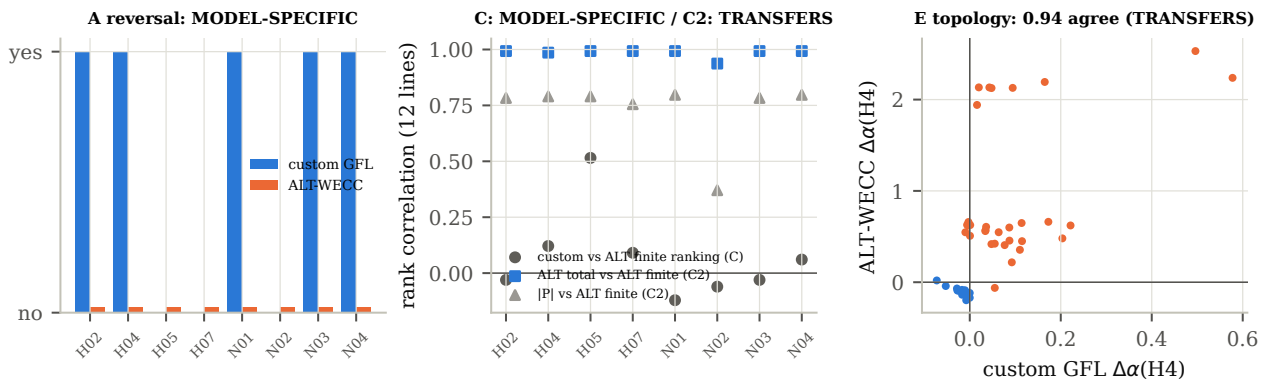


Figure S2: Cross-model holdout (ALT-WECC).

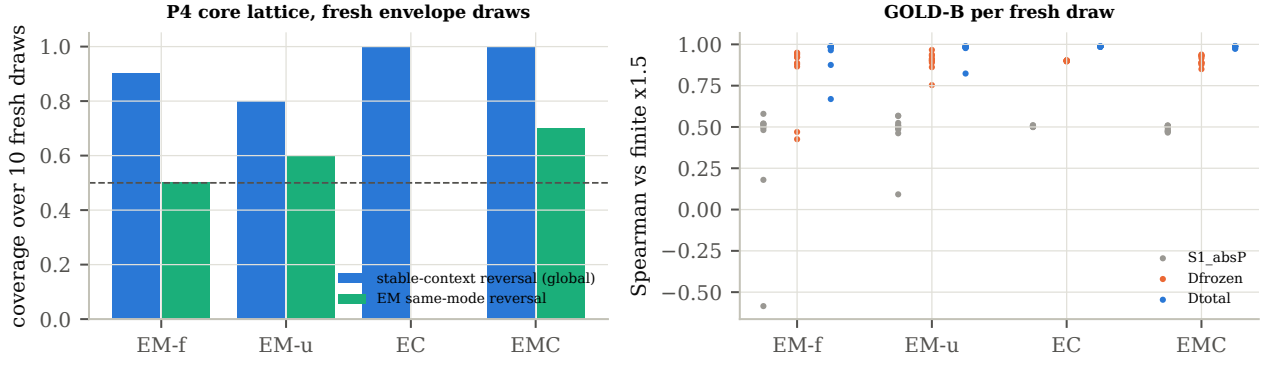


Figure S3: Fresh envelope draws: reversal coverage and branch ranking.

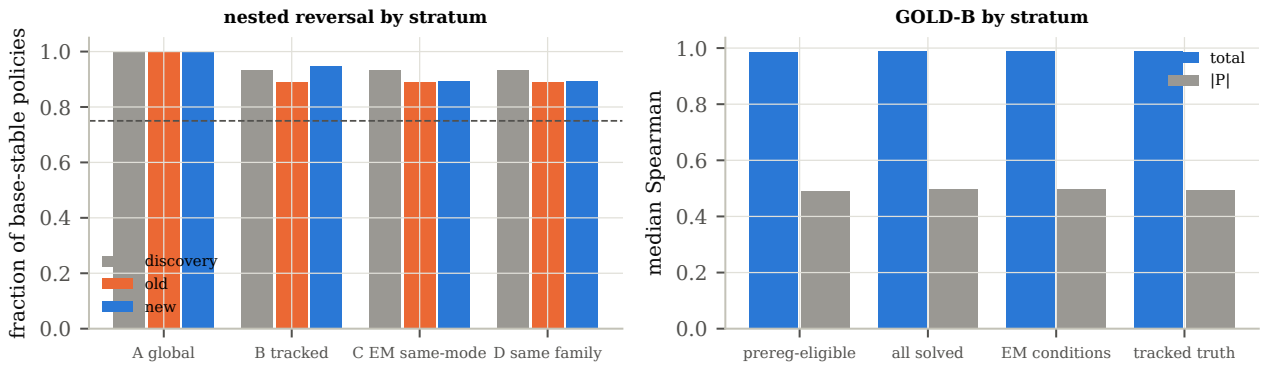


Figure S4: FULL / EM / same-mode stratification.